

CLASS PROJECT V – EMPIRICAL RULE AND Z-SCORES

REVIEW OF KEY CONCEPTS

1. Recap the Empirical Rule (68-95-99.7 Rule):

- The Empirical Rule states that in a normal distribution:
 - 68% of data falls within one standard deviation of the mean.
 - 95% of data falls within two standard deviations.
 - 99.7% of data falls within three standard deviations.
- *Question: Why is this rule useful in understanding data distributions?*

2. Define Z-Scores:

- *What does the Z-score tell us?*
- *How can z-scores help us determine if a data point is typical, unusual, or very unusual in a dataset?*

EXPLORING THE PROVIDED DATASET

1. Create a histogram for each column in the dataset (e.g., Resting Heart Rates, Male Heights, Female Heights, Social Media Usage, Study Hours, Exercise Hours).

2. Calculate Summary Statistics:

- For each column, calculate the mean and standard deviation (generate summary stats)
- Record these values, as they will be **essential** for calculating **z-scores**.

TASKS: COMPARING YOUR OWN DATA WITH THE DATASET

1. Collect Your Own Data:

- Take a few minutes to gather your own data for each variable in the dataset, then ask 4 people in your life for their data:

<i>Names (aliases):</i>					
Resting Heart Rate					
Height					
Social Media Usage(hrs/day)					
Study Hours per Week					
Exercise Hours per Week					

2. Calculate Z-Scores for Your Data:

- Using the mean and standard deviation for each variable (from Part 2), calculate the z-score for each of the data points.

- Formula: z-score = (Your Data - Mean) / Standard Deviation

$$z = \frac{x - \mu}{\sigma}$$

- Interpretation:

- A z-score close to 0 means your data is close to the average.

- A z-score above 2 or below -2 may indicate an unusual value.

Z-scores table

<i>Names (aliases):</i>					
Resting Heart Rate					
Height					
Social Media Usage(hrs/day)					
Study Hours per Week					
Exercise Hours per Week					

3. Compare Your Results:

- Where does each of your data points fall within the distribution?

- Are any of your values “typical” (within ± 2 standard deviation), “unusual” (beyond ± 2 standard deviations), or “very unusual” (beyond ± 3 standard deviations)?

REFLECTION QUESTIONS

How did your personal data compare to the people you asked? Were any of your z-scores surprising?